REED BOARDMAN
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EEG, TMS, performance monitoring, adaptive control, computational neuroscience

EDUCATION

2021 – 2025 **University of Iowa**
Bachelor of Arts in Psychology
Minor in Human Relations

HONORS

2021 - 2025 Honors in Psychology, University of Iowa
2023 – 2025 Dean's List (x4)
2023 – 2025 President's List (x3)
2023 – 2025 Psi Chi Honor Society
2025 Research in the Capitol, Presenter
1 out of 20 Iowa Undergraduate Researchers Selected to Present

RESEARCH EXPERIENCE

Wessel Cognitive Neurology Lab, University of Iowa June 2025 – Present
Post-Baccalaureate Researcher | Advisor: Dr. Jan Wessel
Research Focus: Examining neural networks underlying domain-general inhibitory control, flexible behavior, and cognition.
Ongoing First-Author Projects:

- ***PreSMA Disrupt***
  An rTMS and EEG study investigating the causal role of the pre-supplementary motor area (preSMA) in the 'cancel' stage of inhibitory control using the stop-signal task. We use fMRI-guided continuous theta-burst stimulation (cTBS) in a double-blind, sham-controlled design to disrupt preSMA activity. EMG captures trial-by-trial motor inhibition, isolating early and late inhibitory processes, while EEG characterizes the neural dynamics underlying these effects.
- ***4-GO***
  A novel task dissociating reward acquisition from information gain, enabling estimation of expected value of information and strategic planning under uncertainty. Applying computational modeling to quantify individual differences in information-seeking behavior. Next phase aims to employ 7T fMRI to identify neural substrates exhibiting parametric modulation by distinct model variables. This project forms the basis of my NSF GRFP proposal.

SCAN Lab, University of Iowa November 2023 – June 2025
Part-Time Laboratory Research Assistant | Advisor: Dr. Bruce Bartholow
Research Focus: Examining neurocognitive related to the etiology of alcohol use disorder (AUD)
Independent Thesis: ***Predicting Unique Post-Error Mechanisms of the ERN and Pe Components***

- Collected and analyzed electroencephalogram (EEG) data independently
- Used mixed-effects modeling to probe trial- and subject-level relationships between ERPs and behavior
- Acquired a robust understanding of neurocognitive processes and paradigms
- Disseminated findings to general audiences

Hwang Lab for Neurocognitive Dynamics, University of Iowa August 2024 – May 2025
Undergraduate Research Assistant | Advisor: Dr. Kai Hwang

Research Focus: Investigations into hierarchical cognitive control, thalamocortical contributions to cognition, cognitive flexibility, network mechanisms of control, and integration of task representations.

- Developed a robust foundation in Python and BASH
- Collaborated with graduate students on task and paradigm conception
- Conducted time-frequency decomposition with existing EEG dataset
- Collected behavioral and EEG datasets
- Improved task designs and optimized PsychoPy scripts

Cognitive Control Collaborative (CCC), University of Iowa August 2024 – Present
Member
Collaborating Laboratories: Cockburn, Hazeltine, Hwang, Jiang, and Wessel

- Presented and engaged in biweekly discussions hosted by a rotating schedule of CCC laboratories
- Discussed higher-level questions related to task design, data analysis, and theoretical work
- Gained insights into differing perspectives on cognitive flexibility, task representations, inhibition, control, and neuroimaging methods from each of the laboratories

POSTERS/PUBLICATIONS

1. **Boardman, R.,** Brancaleone, P., Bartholow, B. (2024) *Have Some Cognitive Control: Investigating the Role the ERN and Pe in Error Monitoring and Post-Error Behavioral Adjustments Using Mixed-Effects Modeling.*

2. **Boardman, R.,** Brancaleone, P., Bartholow, B. (2025) *From Voltage to Motor Output: Using Mixed-Effects Modeling to Investigate the Role of Neural Event-Related Signals in Error Monitoring and Cognitive Control.* Poster presented at **Research in the Capitol (RIC)**, Des Moines, IA.

3. **Boardman, R.**, Bartholow, B. (2025) *Late Error-Related Activity as a Computational Proxy for Internal Error Monitoring and Adaptive Control.* Abstract accepted to **Society for Neuroscience Annual Meeting, San Diego, CA.**

RELEVANT COURSEWORK

- Mixed-Effects Modeling
- Biostatistics
- Neuroscience of Executive Functions
- Intro to Behavioral Neuroscience
- Cognitive Psychology
- How the Brain Learns and Why It Makes Mistakes
- Human Information Processing (Lab)
- Statistics
- Research Methods in Psychology I & II
- Pharmacology of Drug Use and Abuse

SKILLS

Statistical/Analytical

- Programming software for analysis/modeling**: R, Python, MATLab (EEGLab)**
- Statistics and computation: **mixed-effects analysis, Bayesian modeling, multivariate and simple regression, ANOVA, and rANOVA**
- Task programming for cognitive experimental design **(PsychoPy, PsychToolbox)**
- Passion for continued education in inferential statistics and Bayesian methods

Neuroimaging/Physiological Imaging

- Development of EEG preprocessing pipelines in **EEGLab** and **Brainvision Analyzer**
- EEG/ERP analysis
- Time-frequency analysis and decoding
- Electromyography (EMG) collection & analysis

Experimental Methods

- Repetitive transcranial magnetic stimulation (rTMS)
- Proficient in designing and executing cognitive neuroscience experiments
- Eyetracking techniques/software for visual attention (**EyeLink**)

LEADERSHIP & SERVICE

Undergraduate Teaching Assistant
Intro to Cognitive Psychology, Fall 2024 | Advisor: Dr. Cathleen Moore

- Supported students through office hours and exam review sessions.
- Reinforced key course concepts and facilitated group discussion preparation.
- Assisted with reframing material to generate optimal learning outcomes

Collegiate Recovery Program (CRP), Student Coordinator

- Organized weekly meetings for students navigating substance use recovery.
- Facilitated group discussions using a harm-reduction framework.
- Promoted academic resilience and peer support within the university setting

Honors Program in Psychology
Completed Spring 2025

- Designed and executed an independent honors project on neural signals of error monitoring.
- Participated in departmental research seminars and received distinction upon completion

AFFILIATIONS & INVOLVEMENT

Society for Neuroscience, Member (May 2025 – Present)

Psi Chi Honor Society, Member (Spring 2023 – Present)

Iowa Neuroscience Club, Member (Spring 2022 – Present)

Cognitive Control Collaborative (CCC), Member (Fall 2024 – Present)

Computational Modeling Club, Member (Fall 2024 – Present)